NSF 26-509: Integrated Data Systems & Services (IDSS)

Program Solicitation

Document Information

Document History

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		U.S. National Science Foundation Directorate for Computer and Information Science and Engineering Office of Advanced Cyberinfrastructure

Full Proposal Deadline(s) (due by 5 p.m. submitting organization's local time):

July 28, 2026

Fourth Tuesday in July, Annually Thereafter

Category II Submissions

July 27, 2027

Fourth Tuesday in July, Annually Thereafter

Category I and III Submissions

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Important Information and Revision Notes

Proposals must be prepared in accordance with the *NSF Proposal & Award Policies & Procedures Guide (PAPPG)*. Use the version of the guide that is in effect on the proposal's due date.

Summary of Program Requirements

General Information

Program Title:

Integrated Data Systems & Services (IDSS)

Synopsis of Program:

The Integrated Data Systems and Services (IDSS) program supports operations-level national-scale cyberinfrastructure systems and services that broadly advance and facilitate open, data-intensive and artificial intelligence-driven science and engineering research, innovation, and education.

Through this solicitation, the IDSS program is accepting proposals for three categories of projects:

- **Category I.** Development, deployment, and operation of novel national-scale integrated data systems and services, which may include interfacing with or leveraging other existing capabilities, systems and services, as appropriate to the project;
- **Category II.** Transition of established smaller scale, regional, pilot, or prototype data-focused systems and services to national-scale production/operational quality/level. This may also include enhancement and expansion of existing national-scale data-focused operational systems and services; and
- **Category III.** Planning grants for future potential development/deployment or transition/enhancement IDSS projects.

NSF and the Office of Advanced Cyberinfrastructure (OAC) have long supported the development of innovative foundational and application-specific cyberinfrastructure resources and systems to address data-intensive research needs at the campus, regional, and community scales, through programs such as Cyberinfrastructure for Sustained Scientific Innovation (CSSI), Campus Cyberinfrastructure (CC*), and other investments. The primary goal of the IDSS program is to support national-scale foundational data cyberinfrastructure that broadly enables data- and artificial intelligence-driven research for many communities. The IDSS program supports foundational transdisciplinary and demonstrably multi-disciplinary projects aimed to broadly impact the science and engineering research and education community. **Projects that aim to primarily benefit a single science discipline, domain, project, or application are not supported.**

It is recommended that prospective PIs contact program officer(s) from the list of Cognizant Program Officers to gain insight about alignment of their project ideas with the priorities of the IDSS program and Office of Advanced Cyberinfrastructure. As part of contacting Cognizant Program Officers, prospective PIs are also encouraged to ascertain that the focus and budget of their proposed work are appropriate for this solicitation.

Expanding Participation in STEM, NSF Priorities, and Gold Standard Science:

NSF prioritizes cutting-edge discovery science and engineering research, advancing technology and innovation, and creating opportunities for all Americans. NSF has established priorities set forth by Congress, the administration and the NSF director to promote NSF's mission. Proposers should review the list of NSF priorities and are encouraged to align their proposals with them, where appropriate. NSF also expects the highest standards of scientific rigor, integrity and adherence to appropriate tenets of Gold Standard Science in proposals, as appropriate for the field of science and research modality.

Cognizant Program Officer(s):

Please note that the following information is current at the time of publishing. See program website for any updates to the points of contact.

- Andrey Kanaev, telephone: (703) 292-2841, email: idss@nsf.gov
- Kevin L. Thompson, telephone: (703) 292-4220, email: idss@nsf.gov
- Marlon Pierce, telephone: (703) 292-7743, email: idss@nsf.gov

Applicable Catalog of Federal Domestic Assistance (CFDA) Number(s):

- 47.070 --- Computer and Information Science and Engineering

Award Information

Anticipated Type of Award: Standard Grant or Cooperative Agreement

Estimated Number of Awards: 3 to 9

The estimated number of awards in each Category is as follows: 1 to 2 (Category I), 1 to 2 (Category II), 1 to 5 (Category III).

The type of award in each category is as follows: Cooperative Agreement (Category I and II), Standard Grant (Category III)

All projects must be submitted as single proposals. **Collaborative research proposals are not permitted and will be returned without review.**

Anticipated Funding Amount: $500,000 to $60,000,000

Proposal Preparation and Submission Instructions

A. Proposal Preparation Instructions

- **Letters of Intent:** Not required
- **Preliminary Proposal Submission:** Not required
- **Full Proposals:**
  - For proposals submitted via Research.gov, *PAPPG* guidelines apply.
  - For proposals submitted via Grants.gov, *NSF Grants.gov Application Guide* guidelines apply.

B. Budgetary Information

- **Cost Sharing Requirements:**

  Inclusion of voluntary committed cost sharing is prohibited.

- **Indirect Cost (F&A) Limitations:**

  Not Applicable

- **Other Budgetary Limitations:**

  Not Applicable

C. Due Dates

- **Full Proposal Deadline(s)** (due by 5 p.m. submitting organization's local time):

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Proposal Review Information Criteria
Merit Review Criteria:

National Science Board approved criteria. Additional merit review criteria apply. Please see the full text of this solicitation for further information.

Award Administration Information
Award Conditions:

Additional award conditions apply. Please see the full text of this solicitation for further information.

Reporting Requirements:

Additional reporting requirements apply. Please see the full text of this solicitation for further information.

I. Introduction

The transformation of research data to knowledge and discovery at national scale requires high-performing data systems and services that are broadly accessible and reliable for the research community. Such systems and services must be functionally well-defined; be able to interface seamlessly with other external systems as appropriate, such as scientific instrumentation, repositories, and computing resources; support the ability of researchers to adhere to scientific openness and reproducibility aims; and be able to evolve with changing research methods and technologies.

A growing number of national efforts and agency initiatives such as the NSF-led interagency National Artificial Intelligence Research Resource (NAIRR) Pilot are actively developing highly performant and interconnected data infrastructure. *Recent federal*- reports[1][2]have emphasized the need for coordinated planning and deployment of operational data infrastructure to broadly support the research and education community.

Through programs such as CSSI, CC*, and other investments, NSF has long supported the development of innovative foundational and application-specific cyberinfrastructure resources and systems to address data-intensive research needs at the campus, regional, and community scales. The primary goal of the IDSS program is to support **national-scale operational** data cyberinfrastructure that broadly advances and enables data- and artificial intelligence-driven research for many communities. Additional goals include:

- defining expectations for design, development and operational performance of data systems that provide critical services to the research and education community;
- enabling research and education communities to maximize the value, use, and reuse of research data;
- increasing awareness and emphasis on addressing the end-to-end data lifecycle, and facilitating the user journey by providing stable and easily usable services for scientific end users; and
- developing the workforce of capable data infrastructure developers and managers.

II. Program Description

Scope. The IDSS program supports national-scale performant operational systems and services that broadly facilitate open, data-intensive and artificial intelligence-driven science and engineering research, innovation, and education. IDSS projects should be aimed to broadly impact the science and engineering research and education community in a transdisciplinary and demonstrably multi-disciplinary way, enabling researchers and educators from diverse domains and disciplines to utilize research data, integrate data, and connect data sources with other scientific resources such as computing resources, facilities, instrumentation and repositories. **Projects that aim to primarily benefit a single science discipline, domain, project, or application are not supported.**

Emphasis on integration. The IDSS program supports projects that demonstrably contribute to the vision of an integrated, federated and accessible advanced research cyberinfrastructure ecosystem that meets the Nation's foundational needs for world-leading data, computing, and networking capabilities. Projects are expected to leverage and interconnect with other existing operational cyberinfrastructure systems and services and other data and relevant facilities, whether supported by NSF or by other entities, as appropriate to project objectives. Inline with this emphasis, **all proposed projects, including collaborative projects, must be submitted as a single proposal in which a single award is being requested** (PAPPG Chapter II.E.3.a). The involvement of partner organizations should be supported through subawards administered by the submitting organization.

Scientific data lifecycle. The IDSS program aims to develop a portfolio of projects that collectively enable data utilization pathways and workflows across the end-to-end scientific data lifecycle. The IDSS program has an inclusive and flexible view of the scientific data lifecycle that may include stages and functionalities such as acquisition, transfer, management, exploration, analysis, curation, sharing, synthesis, discovery, and archiving, as may be defined by a project or community. A given IDSS project need not support all stages of a reference data lifecycle but must be clear about how the project enables one or more scientific pathways through all or few stages of a lifecycle.

Storage and curation. The IDSS program supports integrated resources, services and environments to enable hosting, manipulation of, and workflows for research data. **The IDSS program does not support costs for permanent long-term hosting, storage, archival, and curation of the research data itself.** Projects that involve partnerships, fee-based models, or other such mechanisms to support these long-term data storage and curation costs are encouraged.

Innovation and adaptability: A portion of an IDSS project is expected to be dedicated to innovation and improvement of operational services over the lifetime of the award. Proposed projects designed to enable research communities to build customized tools and capabilities upon the IDSS-supported project infrastructure are also encouraged. **Projects that have the goal of cyberinfrastructure innovation without operations expectations and plans are not supported.**

Relationship to other funding programs. Proposed IDSS projects should not be appropriate for funding by any other current NSF programs or solicitations. The IDSS program is complementary to other production/operations-oriented national-scale cyberinfrastructure programs supported by the Office of Advanced Cyberinfrastructure (OAC) including the Advanced Systems and Services Program (ACSS) and the ACCESS coordinated services program. ACSS and ACCESS address advanced computing needs of the broad S&E community; the IDSS program focuses on data infrastructure. The IDSS program is also complementary to the OAC CSSI and CC* programs. CC* emphasizes institutional and regional capabilities and CSSI primarily supports data and software infrastructure development; IDSS supports national-scale operational projects. Prospective proposers of pilot- and prototype-stage projects should consider other OAC programs such as CSSI or other relevant NSF programs.

Programmatic areas of interest

Current areas of particular interest to the IDSS program include, but are not limited to and may involve a combination of:

- Projects that facilitate the connection of data sources with advanced computing resources and analytic environments in integrative ways for an appropriately broad array of use cases.
- Projects that address the emerging data-intensive workflows and data integration needs of artificial-intelligence (AI)-driven research (including research about AI and research using AI capabilities).
- Projects that focus on enabling one or more specific points in the data lifecycle applied at a national scale.

Projects that enhance the ability of the research and education community to access and utilize open research data supported by other federal agencies are welcome, provided that such projects are not primarily benefitting a single science discipline, domain, project, or application, and are complementary to, not overlapping with investments being made by those other agencies for similar purposes.

Proposal categories and descriptions

IDSS offers the following three categories of proposals:

- **Category I.** Development, deployment, and operation of novel national-scale integrated data systems and services, which may include interfacing with or leveraging other existing capabilities, systems and services as appropriate to the project.
  - Between $10 million to $30 million for up to 5 years. Potentially renewable.
- **Category II.** Transition of established smaller scale, regional, pilot, or prototype data-focused systems and services to national-scale production/operational quality/level. This may also include enhancement and expansion of existing national-scale data-focused operational systems and services.
  - Up to $9 million for up to 3 years. Potentially renewable.
- **Category III.** Planning grants for future potential development/deployment or transition/enhancement IDSS projects.
  - Up to $500,000 for up to 2 years. Not renewable.

Category I projects. Projects proposed in this category are intended to result in deployment and operation of novel national-scale data-focused systems and services to support a broad range of data-intensive and AI-driven research use cases. **Proposals that are effectively continuations of existing data infrastructure projects with minimal changes are not supported and may be returned without review.** The proposed data systems and services must be clearly motivated by identified AI and/or other use cases representing the broad disciplinary and geographically diverse research and education communities, and analysis of current and future demand for associated data usage and utilization patterns. Projects are expected to leverage and interconnect with other existing operational cyberinfrastructure systems and services and other data and relevant facilities, whether supported by NSF or by other entities, as appropriate to project objectives. Projects are also encouraged to explore novel models for future dynamic national cyberinfrastructure federation.

Category II projects. Projects proposed in this category have similar expectations as Category I projects and are intended to result in similar outcomes as Category I projects. While Category I projects may result in entirely new systems and services or compositions of systems and services, Category II projects are expected to build upon an established existing performant project with defined capabilities, systems and services, which may include pilots and prototypes, that are demonstrably successful at serving an array of users at a smaller-than-national scale and which seeks to scale up to a national operational infrastructure. Category II projects may also include targeted enhancements and expansions of existing national-scale data-focused operational systems and services.

Category III projects. Projects proposed in this category are intended as planning grants to develop plans and prepare for proposing future Category I or Category II IDSS proposals. Category III proposals cannot include any technical development or operational activities. Activities to engage targeted research/user communities are supported and encouraged as part of a planning effort.

The source code of any software developed under an IDSS project must be made publicly available and assigned an identified open-source software license.

Potential for renewal awards.

In Category I or Category II, there is a possibility of a renewal award contingent upon availability of funds and the successful evaluation of the project's performance as well as NSF merit review of the renewal proposal. Contingent on a successful third-year review, a Category I project may be invited by NSF to submit a renewal proposal for up to five years commencing at the conclusion of the original award. Contingent on a successful second-year review, a Category II project may be invited by NSF to submit a renewal proposal focused on operations and innovation during operations for up to five years commencing at the conclusion of the original award. Note that renewal proposals will need to be developed as fully as though the proposer were applying for the first time and should cover all the information required in a proposal for a new project, including Results from Prior NSF Support. Category III awards are not eligible for renewal.

III. Award Information

Estimated program budget, number of awards and average award size/duration are subject to the availability of funds.

IDSS awards will be supported at the following budget levels and durations:

- Category I awards: Between $10 million to $30 million for up to 5 years.
- Category II awards: Up to $9 million for up to 3 years.
- Category III awards: Up to $500,000 for up to 2 years.

In Category I or Category II, there is a possibility of a renewal award contingent upon availability of funds and the successful evaluation of the project's performance as well as NSF merit review of the renewal proposal. Category III awards will not be eligible for renewal.

IV. Eligibility Information

Who May Submit Proposals:

Proposals may only be submitted by the following:

- Institutions of Higher Education (IHEs): Two- and four-year IHEs (including community colleges) accredited in, and having a campus located in the US, acting on behalf of their faculty members. Special Instructions for International Branch Campuses of US IHEs: If the proposal includes funding to be provided to an international branch campus of a US institution of higher education (including through use of sub-awards and consultant arrangements), the proposer must explain the benefit(s) to the project of performance at the international branch campus, and justify why the project activities cannot be performed at the US campus.
- Non-profit, non-academic organizations: Independent museums, observatories, research laboratories, professional societies and similar organizations located in the U.S. that are directly associated with educational or research activities.
- Other Federal Agencies and Federally Funded Research and Development Centers (FFRDCs): Prospective proposers from other Federal Agencies and FFRDCs, including NSF sponsored FFRDCs, must follow the guidance in PAPPG Chapter I.E.2 regarding limitations on eligibility.

Who May Serve as PI:

There are no restrictions or limits.

Limit on Number of Proposals per Organization: 1

An organization may submit only one proposal as lead institution for each of Category I and Category II for each solicitation deadline but may be a subawardee on other Category I and II proposals responding to this solicitation. The restriction to no more than one submitted proposal as lead institution is to help ensure that there is appropriate institutional commitment necessary for responsible oversight, by the potential recipient institution, of a national data infrastructure resource. This restriction does not apply to Category III proposals.

In the event that any organization exceeds this limit, any proposal submitted to this solicitation from an organization after the first proposal is received at NSF will be returned without review. No exceptions will be made.

Category III. There are no restrictions or limits.

Limit on Number of Proposals per PI or co-PI: 1

An individual may participate as PI, co-PI, or other Senior Personnel on at most one proposal across Categories I and II for each solicitation deadline. Thus, if an individual participates on a Category I proposal, the individual may not participate in a Category II proposal, or vice versa. Note that any individual whose biographical sketch is provided as part of the proposal will be considered as Senior Personnel in the proposed activity, with or without financial support from the project. Other staff who will be funded under an IDSS proposal are not subject to this restriction.

This restriction does not apply to Category III proposals.

In the event that any individual exceeds this limit, any proposal submitted to this solicitation with this individual listed as PI, co-PI, or Senior Personnel after the first proposal is received at NSF will be returned without review. No exceptions will be made.

V. Proposal Preparation and Submission Instructions

A. Proposal Preparation Instructions

Full Proposal Preparation Instructions: Proposers may opt to submit proposals in response to this Program Solicitation via Research.gov or Grants.gov.

You must prepare your proposal according to Chapter II.D.2 of the PAPPG, unless this solicitation specifies different instructions. Always use the version of the PAPPG in effect on your proposal's due date.

- For proposals submitted via Research.gov, *PAPPG* guidelines apply.
- For proposals submitted via Grants.gov, *NSF Grants.gov Application Guide* guidelines apply.

In determining which method to utilize in the electronic preparation and submission of the proposal, please note the following:

Collaborative Proposals. All collaborative proposals submitted as separate submissions from multiple organizations must be submitted via Research.gov. PAPPG Chapter II.E.3 provides additional information on collaborative proposals.

The following information supplements the guidelines and requirements in the NSF PAPPG and NSF Grants.gov Application Guide:

1. Cover Sheet:

- **Proposal Title:** Title must begin with either "Category I: ", "Category II: ", or "Category III: " as the case may be, followed by the title.
- **International Partners:** If your project involves international partners, check the international activities box and list the countries involved. In keeping with NSF typical practice, the IDSS program does not permit awards to include funding for international partners.
- The NSF proposal system allows one PI and at most four co-PIs to be designated for each proposal. If needed, additional lead personnel should be designated as non-co-PI Senior Personnel on the Budget form.
- Collaborative efforts may only be submitted as a single proposal (See PAPPG Chapter II.E.3.a), in which a single award is being requested. The involvement of partner organizations should be supported through subawards administered by the proposing Resource Provider organization.

2. Project Summary

3. Project Description:

The page limit for the Project Description section of the proposal is **20 pages** for Category I and Category II proposals and **10 pages** for Category III proposals.

3a. Project Description Guidance for Category I and II proposals and projects:

In addition to the guidance specified in the PAPPG, including a separate section labeled Broader Impacts, and in alignment with the description of the Category I and Category II proposal categories above, the Project Description must include the following sections:

1. Vision, goals, and driving requirements
2. Project definition and specification
3. Concept of Operations
4. Performance objectives and measures
5. Project management
6. Budget estimation

(1) Vision, goals, and driving requirements

Proposals must articulate a clear vision and concise goals for project impacts on science and innovation, including but not limited to growing, onboarding, and serving a broad array of users. Projects must be ultimately aimed to enable and accelerate open scientific discovery and innovation by the U.S. research and education community, and materially and demonstrably advance the capabilities, capacity, and operational interconnectedness of the nation's research cyberinfrastructure ecosystem.

Proposals must identify and discuss **specific reference research, innovation and/or education use cases** that require the capabilities of the proposed cyberinfrastructure, any community and national-scale reports and other such evidence of need for the proposed systems and services, and the kinds of science outcomes that would be enabled or accelerated.

Based on this evidence of need, proposals must identify the specific requirements the design, performance and operational plans for achieving the vision and goals for the proposed data infrastructure systems and services.

Category II proposals must, in this section, qualitatively and quantitatively demonstrate the demand and user base for their existing systems and services and discuss the current performance objectives and associated measures, and actual performance of the existing project.

(2) Project definition and specification

Proposals must enumerate and describe the proposed project's architecture and intended systems and services and how these each respond to the vision, goals and requirements described in section (1) above. Proposals must also identify and/or describe

- the types or categories of intended users to be supported, appropriate to a broadly accessible national-scale system;
- an end-to-end scientific data lifecycle model or process that will be adopted as a reference, and identify which lifecycle stages, and utilization patterns and workflows, will be primarily supported;
- how the project will leverage and connect, as appropriate, with other existing operational cyberinfrastructure, data sources, software systems, and other facilities and platforms, as appropriate to achieving the project goals; and
- uniqueness, complementarity, and potential synergies of the proposed project with other deployed cyberinfrastructure systems, platforms and services, and must make a compelling case for a unique fit or advantages of the proposed project relative to other alternatives to enhance the NSF cyberinfrastructure ecosystem.

(3) Operations Plan

IDSS Category I and Category II awards support development/deployment and operations. Projects will be expected to achieve readiness for full operations **by or before the end of the first award year,** after which the project will commence continuous operations of all systems and services as measured against defined performance targets. Acceptance of the project and transition to full operations may be subject to NSF-conducted review and approval, as appropriate to the project and to be defined in the award terms and conditions.

Proposals must include a clear plan for operation and maintenance of the systems and services for the duration of the award period, including at a minimum but not limited to:

- functionalities and modes of operation as well as utilization and associated activities;
- activities related to user engagement, onboarding, robust support and training at a level commensurate with national-scale systems and services to be provided;
- number, types and anticipated qualifications of the project personnel that will be involved;
- as appropriate to the project, the plan for how the proposed resources and services will be apportioned, allocated, and/or otherwise made accessible to end users, for what time periods, and incorporating principles of broad, open, merit-based, access;
- the plan for early user access to the systems and services that will be provided during the early operations and testing phase(s)prior to project acceptance and how this early utilization will inform final operations plans;
- anticipated technology refreshes, upgrades and innovations, and outlining the strategy for these; and,
- any other operational details likely to have an impact on user access or usage of the proposed systems and services.

(4) Performance objectives and measures

Proposals must provisionally define a comprehensive set of objectives and targets as appropriate for the system and services proposed and discuss how these will be achieved. Such objectives might include targets for end-to-end performance, user wait times, interactivity and system responsiveness, growth in user community, stability, reliability, or usability targets.

Proposals must also describe how the project will be measured qualitatively and quantitatively to sufficiently enable assessment of project performance against these objectives and targets and the vision and goals. This should include demonstrating user satisfaction with the systems and services and the impact on scientific research and innovation.

Projects will be expected to openly expose the usage of the system in ways that are appropriate to the project and its goals.

(5) Project management

This section of the proposal should be organized according to the following subsections and headings.

i. Project leadership team and their experience. Strong project organization and management capabilities and experience must be demonstrated in the proposal. The IDSS program expects that the PI or a dedicated Project Leader or Project Manager with leadership authority for the project, and other management team members, will be devoting substantial portions of their time to the project. Proposals must describe the experience of the proposing organization and team in managing awards for development and operation of cyberinfrastructure systems of similar scale and complexity to that being presently proposed that have a potentially large number of external users, and the resources that would be available to manage the new award. This should include, as appropriate, indication of experience with providing continuous operations, the number and types of users, and the nature of the user support provided. Proposals should also describe any plans to draw on external experts to help advise and address any system development, integration, and operational challenges that arise during the project.

ii. Project organization and timeline. Proposals must clearly lay out the organization of the project activities to be performed during development and operations. This must include, at a minimum,

   - Description of the specific roles and responsibilities of the PI, co-PIs, other senior personnel, and paid consultants at all involved organizations in management and execution of the project;
   - Project organization in terms of the development work to be performed, with associated named individuals or positions and budgets for each major category (such as a work breakdown structure to level 2 or 3).
   - The temporal stages of activity in the evolution of the project, for example, development, deployment, early operations, and full operations, as appropriate;
   - High-level project development schedule by quarters (or month-by-month, as appropriate), and indicating key milestones and associated dates for project start, transition(s) between the major phases (which may include overlap of phases, as appropriate) including start of full operations; and,
   - A plan including the described process for demonstrating, validating, and confirming the performance of the proposed systems and services in an early operations or equivalent phase ahead of full operations.

   If managing roles and responsibilities for project execution will be distributed across multiple organizations, proposals must discuss how this arrangement will result in sufficient and unified accountability, reporting chain, and decision-making authorities to achieve successful execution of the project within the proposed budget and schedule.

iii. Outsourcing. Proposals must provide details and describe the contractual terms for any planned vendor and other sub-contracts for substantial acquisitions or other outsourcing of hardware, software, or services, and describe the experience and organizational resources of the proposing organization and team for the management and oversight of such sub-contracts at this scale. Any planned service level agreements (SLAs) should be described in this section in measurable and verifiable terms.

iv. Software. Proposals must describe any major software components of the project and how these will be either developed or otherwise obtained. If the proposed project includes in-house development of software components, the proposal must describe and demonstrate how the experience of the proposing team and organizational resources are adequate for these activities as part of deploying and operating a national scale infrastructure. If the project involves acquisition of software licenses or services, the proposal should provide appropriate justification and comparison with other open-source solutions or in-house development efforts.

v. Risk management. Proposals must provide a detailed risk management plan, identifying the major technical and management risks to the project, including risks associated with community adoption and sustainability, and associated planned mitigations, for successfully completing development/deployment and conducting full operations in the planned time period within the proposed budget.

vi. Security and trustworthiness. Proposals must describe how the project will implement robust physical and cybersecurity measures, as appropriate to the project, and how these are aligned with the project's accessibility, trustworthiness, and usability aims. This should include the reference policies and procedural regimes to be adopted, project roles and responsibilities, how related risks will be assessed, technical and administrative safeguards, plans for awareness and training, and procedures for notification to NSF and the relevant user and cyberinfrastructure communities, and appropriate authorities. Proposers should also describe how the effectiveness of the security program will be evaluated and assessed.

If awarded, the Project PIs will be required to develop a Project Execution Plan (PEP) for NSF review and concurrence. Regarding schedule planning, proposers should plan to be able to proceed with any long-lead acquisitions and urgent development work in the awarded project while the full PEP is in development. A detailed risk-mitigated deployment plan will be required as part of the PEP to ensure that the proposed systems and services will enter into full operations and be available to the user community within the required timeframe.

(6) Budget estimation

Proposals must include a detailed cost estimate for all planned activities and costs across the duration of the award as a Supplementary Document. This section of the Project Description should summarize the cost estimate at a high level.

Unallowable costs. Proposers should note that the IDSS program will not provide support for the following activities:

- Renovation of buildings or labs to accommodate the infrastructure
- Individual research enabled by the infrastructure.

Other funding sources. In this section, proposals must also identify any other sources of funding for the project being proposed, and whether any of the scope of current work would continue to receive that support during the proposed IDSS-supported period, as well as how the proposers will ensure that such scope does not overlap with that being proposed to be supported by IDSS. For Category II projects, proposers must provide a plan for concluding any current award supporting the related current activities and transition to the IDSS-supported project.

Proposals should budget for participation in an annual PI meeting or major cyberinfrastructure conference with travel costs supported by the award, to be confirmed with the cognizant Program Director if the project is awarded.

3b. Project Description Guidance for Category III proposals

IDSS Program planning grants are intended to support individuals or groups to improve their readiness for future submission of an IDSS Category I or Category II proposal. Proposals must describe the specific cyberinfrastructure project concept that will be the focus of the planning activities, the vision, goals, and envisaged impact for that eventual project, and the specific activities to be undertaken. Planning projects are strongly encouraged to include significant community engagement to determine and confirm community needs, priorities, and support for the eventual infrastructure and to provide input into the concept development. Ideally, Category III projects will produce robust conceptual plans for the development of a national scale cyberinfrastructure project, taking into consideration the expectations listed in this solicitation for Category I and/or Category II proposals, as appropriate.

Proposals should budget for participation in an annual PI meeting or major cyberinfrastructure conference with travel costs supported by the award, to be confirmed with the cognizant Program Director if the project is awarded.

Broader Impacts (applies to all proposal categories)

In addition to the instructions provided in the PAPPG, proposals must describe any complementary and leveraged aspects within the CI ecosystem, with emphasis on other NSF-funded CI projects and priorities described in the Program Description. Proposals should clearly identify the broader impacts of the project, such as benefits to science and engineering communities beyond initial targets, and education and workforce development.

Supplementary Documents:

In addition to the guidance specified in the PAPPG, the following Supplementary Documents should be included. Proposals missing any of the required documents will be **returned without review.**

- Only for Category I and Category II proposals: Proposals must include a supplementary document containing a detailed cost estimate for all planned activities and costs across the duration of the award. The cost estimate should be organized by project organizational elements and project phases identified in the Project Management section of the Project Description (V.A.3a.(5)). The cost estimate should include, as appropriate, estimates for resource consumption (power, etc.) and connecting to external resources (e.g., network connectivity, connections to/services leveraged from other resources), and any necessary maintenance contracts, outsourcing, and licenses. The basis of estimates for each of the major activities or cost areas should be identified and discussed, and explanations of labor costs should identify the number, types, and salaries of full-time equivalent (FTE) staff, particularly including those required for development, maintaining continuous operations and user support, training, and outreach. Any other factors that are anticipated to have an impact on the Total Cost of Ownership of the proposed resource, as appropriate, must also be provided.

Letters of Collaboration (optional): Include documentation of funded or unfunded collaborative arrangements of significance to the proposal through Letters of Collaboration. [See PAPPG Chapter II.C.2.j for details.] Letters of Collaboration should be limited to stating the intent to collaborate and should not contain endorsements or evaluation of the proposed project. The recommended format for Letters of Collaboration is as follows:

"If the proposal submitted by Dr. [insert the full name of the Principal Investigator] entitled [insert the proposal title] is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal."

Scan your signed letters of collaboration, containing only text similar to the above, and upload them into the Supplementary Documents section of the proposal, but do not send originals.

Do not submit letters of support. For example, letters of endorsement and letters of a laudatory nature for the proposed project are not acceptable.

- ***Project Personnel and Partner Organizations*** (required for all award categories): Provide current, accurate information for all personnel and organizations involved in the project. The list must include all PIs, co-PIs, Senior Personnel, paid/unpaid Consultants or Collaborators, Subawardees, Postdocs, project-level advisory committee members, and writers of letters of support. This list should be numbered and include (in this order) Full name, Organization(s), and Role in the project, with each item separated by a semi-colon. Each person listed should start a new numbered line. For example:
- Mei Lin; XYZ University; PI
- Jak Jabes; University of PQR; Senior Personnel
- Jane Brown; XYZ University; Postdoctoral Researcher
- Rakel Ademas; ABC Inc.; Paid Consultant
- Maria Wan; Welldone Institution; Unpaid Collaborator
- Rimon Greene; ZZZ University; Subawardee

Single Copy Documents:

Collaborators and Other Affiliations Information: Proposers should follow the guidance specified in Chapter II.C.1.e of the NSF PAPPG.

Note the distinction to item on "Project Personnel and Partner Organizations", above, for Supplementary Documents: the listing of all project participants is collected by the project lead and entered as a Supplementary Document, which is then automatically included with all proposals in a project. The Collaborators and Other Affiliations are entered for each participant within each proposal and, as Single Copy Documents, are available only to NSF staff.

No other items or appendices are to be included. Full proposals containing items other than those required above or by the Proposal and Award Policies and Procedures Guide (PAPPG) will be **returned without review.**

B. Budgetary Information

Cost Sharing:

Inclusion of voluntary committed cost sharing is prohibited.

Budget Preparation Instructions:

Awardees are expected to participate in annual PI meetings with travel costs supported by the award. These travel costs should be included in the proposed budget. Proposals with total project budgets exceeding the maximum amount allowed in the proposal Category will be **returned without review.**

Proposers are reminded that the IDSS program will not provide support for the following activities:

- Renovation of buildings or labs to accommodate the infrastructure
- Individual research enabled by the infrastructure.

B. Budgetary Information
Cost Sharing:

Inclusion of voluntary committed cost sharing is prohibited.

C. Research.gov/Grants.gov Requirements
You can submit proposals in response to this solicitation through Research.gov or Grants.gov, unless otherwise noted.

Information on how to prepare and submit proposals is available on the Submitting Your Proposal page on NSF.gov.

VI. NSF Proposal Processing and Review Procedures

Information on NSF's proposal processing and review procedures is available on the Overview of the NSF Proposal and Award Process page on NSF.gov.

A. Merit Review Principles and Criteria

All NSF proposals are evaluated through use of the two National Science Board-approved merit review criteria:

- **Intellectual Merit**, which encompasses the potential to advance knowledge.
- **Broader Impacts**, which encompass the potential to benefit society and contribute to the achievement of specific, desired societal outcomes.

Information on NSF's merit review principles and process can be found on the How We Make Funding Decisions page on NSF.gov.

Additional Solicitation Specific Review Criteria

Additional merit review criteria apply. Please see the full text of this solicitation for further information.

For Category I and Category II proposals, reviewers will be asked to use the following guiding questions to assess the adequacy of the information provided in the required sections of the Project Description and supplementary documents (as described in Section V.A.3a, Proposal Preparation Instructions above):

- Does the proposal provide convincing evidence that the project is needed, relative to other existing systems and services, to advance integrative data-intensive and AI-driven science and engineering research and education, in integrative ways for an appropriately broad array of use cases?

- Does the proposal provide a clear definition of the proposed systems and services and provide the technical specifications at an appropriate level of detail?
- Is there a convincing plan for transitioning the project to full operations by the end of the first award year?
- Does the Operations Plan clearly describe how the systems and services will provide continuous operations to the intended users while ensuring security and trustworthiness?
- Does the proposal describe an adequate and convincing set of comprehensive performance objectives and appropriate targets for the proposed system or services?
- Are the project management plans, risk management plan, and budget well-described and appropriate?

For Category III Planning proposals:

- Would the envisaged eventual Category I or Category II project be of a stature and impact to advance the goals of the IDSS program "to support **national-scale operational** data cyberinfrastructure that broadly advances and enables data- and artificial intelligence-driven research for many communities"?
- Does the proposal provide evidence that the eventual Category I or Category II project would address transdisciplinary data cyberinfrastructure needs to address data-intensive workflows and data integration needs of artificial-intelligence (AI)-driven research?
- Is there a well-designed planning process and set of activities, including engaging the relevant communities, sufficient to inform sound designs and project plans to develop an eventual competitive IDSS Category I or Category II proposal?

B. Review and Selection Process

Proposals submitted in response to this program solicitation will be reviewed by

Ad hoc Review and/or Panel Review, or Reverse Site Review.

Proposals submitted in response to this program solicitation will be reviewed by Ad hoc Review and/or Panel Review and possibly followed by a Reverse Site Visit.

After a proposal passes an initial compliance check, it will be reviewed by an NSF Program Officer. In most cases, three or more external experts will also review it (either as ad hoc reviewers, panelists or both).

Visit the Overview of the NSF Proposal and Award Process page for more information on the proposal review and selection process.

VII. Award Administration Information

A. Notification of the Award

Notification of an award is made to *the submitting organization* by an NSF Grants and Agreements Officer.

B. Award Conditions

Information on NSF award conditions can be found on the Award Terms and Conditions page on NSF.gov and Chapter VII of the PAPPG.

Administrative and National Policy Requirements:
Information on administrative and national policy requirements can be found on the National Policy Requirements for Recipients of NSF Awards page on NSF.gov.

Special Award Conditions:

- Awardees should budget to participate in an annual PI meeting or major cyberinfrastructure conference with travel costs supported by the award, to be confirmed with the cognizant Program Director.

C. Reporting Requirements

Unless your award notice says otherwise, NSF requires the principal investigator of every grant to submit annual project reports and a project outcomes report for the general public. For complete reporting requirements, see Chapter VII of the PAPPG.

Additional reporting requirements will be negotiated with the recipient prior to award and will be incorporated into the special terms and conditions of the award. Such requirements may include, for example, monthly and quarterly reports reverse-/site visits, and other requirements to enable NSF oversight of the award. The level of oversight will be appropriate to the complexity of the award.

VIII. Agency Contacts

For questions related to the use of NSF systems contact:

- **Research.gov**: NSF IT Service Desk at rgov@nsf.gov or 1-800-381-1532. The Service Desk is open from 7 a.m. to 9 p.m. Eastern time, Monday through Friday (except for federal holidays).

For questions relating to Grants.gov contact:

- **Grants.gov**: The Grants.gov Contact Center at support@grants.gov or 1-800-518-4726. (Contact if the Authorized Organizational Representative (AOR) has not received a confirmation message from Grants.gov within 48 hours of submitting an application.)

IX. Other Information

For information on NSF directorates, programs and funding opportunities, go to NSF.gov.

About the National Science Foundation

The U.S. National Science Foundation is an independent federal agency created by the "National Science Foundation Act of 1950." More information about NSF can be found on NSF.gov.

- **Location:** Randolph Building, 401 Dulany Street, Alexandria, VA 22314
- **General Information** (703) 292-5111
- **TDD (for the hearing-impaired):** (703) 292-5090

Privacy Act and Public Burden Statements

The information requested on proposal forms and project reports is solicited under the authority of the "National Science Foundation Act of 1950," as amended. More information can be found on the Privacy Act and Public Burden Statements page on NSF.gov.

An agency may not conduct or sponsor, and a person is not required to respond to, an information collection unless it displays a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 3145-0023. Public reporting burden for this collection of information is estimated to average 12 hours per response, including the time for reviewing instructions. Send comments regarding the burden estimate and any other aspect of this collection of information, including suggestions for reducing this burden, to:

Suzanne H. Plimpton
Reports Clearance Officer
Office of the Director
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401 Dulany Street
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